

Practice Quiz: Learning to Invent

Part A: Multiple Choice

1. In Active Inference, parameter learning for inventors involves: A) Constructing entirely new frameworks for understanding problems B) Adjusting values within an existing model to improve accuracy C) Learning new physical skills like soldering or machining D) Abandoning the current model and starting over
2. Structure learning differs from parameter learning in that: A) It is faster and requires less effort B) It changes the model itself rather than adjusting values within the existing model C) It only applies to mathematical models, not physical inventions D) It is impossible for experienced inventors
3. Deliberate practice for invention is characterized by: A) Repeating comfortable activities until they become automatic B) Practicing only on real invention projects, never on exercises C) Targeted exercises addressing specific weaknesses, with feedback D) Spending equal time on all skill areas regardless of proficiency
4. When reading a patent as an Active Inference artifact, the "Background of the Invention" section corresponds to: A) The updated generative model incorporating the invention B) The prediction error that motivated the invention C) The prior generative model — the state of knowledge before the invention D) The experimental validation of the invention
5. Biomimicry is most effective when it transfers: A) The exact morphology (shape) of biological organisms to engineering B) Functional principles (structural relationships) rather than surface forms C) Only ideas from organisms closely related to the engineering domain D) Complete biological systems without modification
6. The "learning spiral" in inventive practice consists of: A) Reading, writing, presenting, and publishing B) Perception, cognition, action, and learning — cycling repeatedly with each cycle building on the last C) Planning, executing, measuring, and reporting D) Ideation, prototyping, testing, and launching

7. Edison's systematic approach to inventive learning included: A) Waiting for inspiration to strike before beginning work B) Working exclusively on one invention at a time until completion C) Daily experiment quotas, detailed notebooks, and systematic extraction of principles from both successes and failures D) Hiring the best experts and delegating all experimental work

Part B: Short Answer / Analysis

1. An inventor has spent two years refining their expertise in 3D printing for product prototyping. They are excellent at creating detailed prototypes but consistently struggle with identifying what to prototype — their problem framing and need-finding skills are weak. Using the three modes of learning (parameter, structure, policy), diagnose their learning imbalance. What type of learning have they been doing, what type have they neglected, and what specific deliberate practice activities would address the gap?
2. The Eastgate Centre in Harare was designed by learning from termite mound thermodynamics. The architect had to translate a biological principle (passive ventilation via convection) into an architectural specification (chimney design, wall thickness, opening placement). Analyze this translation process using Active Inference. What did the architect need to learn beyond what the biological analogy directly provided? What was the role of the learning spiral (perception of termite mounds, cognition about thermodynamics, action in building scale models, learning from thermal performance data)?
3. You want to improve your inventive skills over the next three months. Design a structured learning plan that includes: (a) one parameter learning activity (improving accuracy within your current domain knowledge), (b) one structure learning activity (developing a fundamentally new way of thinking about your invention problem), and (c) one policy learning activity (acquiring a new hands-on skill). For each activity, specify what you will do, how often, how you will measure progress, and how it connects to your current invention project.